Supriya Nayak

PRIVACY-PRESERVING WI-FI CSI-BASED FALL DETECTION SYSTE

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FORM 2

THE PATENTS ACT, 1970

(39 of 1970)

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PROVISIONAL SPECIFICATION

(See section 10 and rule 13)

1. TITLE OF THE INVENTION

PRIVACY-PRESERVING WI-FI CSI-BASED FALL DETECTION SYSTEM WITH EDGE NEURAL INFERENCE AND EVENT-DRIVEN AGENTIC VOICE INTERACTION FOR RULE-BASED CAREGIVER ESCALATION

2. APPLICANT

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3. PREAMBLE

The following specification describes the invention.

4. DESCRIPTION

4.1 FIELD OF THE INVENTION

The present invention relates to Wi-Fi-based fall detection and monitoring of elderly persons. It provides a contactless system that detects falls using Wi-Fi channel state information (CSI), processes the data using a compact neural network on an edge device, and initiates a voice interaction with the person after a detected event. The system interprets the person's response using a language model, while a separate rule-based safety controller determines whether to notify a caregiver.

4.2 BACKGROUND OF THE INVENTION

Falls are a major cause of injury among elderly people living alone. Existing fall-detection systems can be divided into three main categories:

1. **Wearable devices** (accelerometer pendants and watches) must be worn and charged regularly. They may be left unused, particularly by people with cognitive or motor impairments.
2. **Camera-based systems** raise privacy concerns and may not be suitable for private areas such as bathrooms and bedrooms.
3. **Wi-Fi sensing systems** avoid these issues, and several patent documents on fall detection are known [1] to [4].

Systems that interact with a person after detecting an event often rely on fixed scripts or keyword matching. These methods may not recognise different ways of expressing the same response, especially when different languages are used. On the other hand, relying only on a generative language model to decide whether to raise an emergency alert may lead to inconsistent decisions. Cloud-based speech and language processing may also raise concerns about privacy, response time, and network connectivity.

Many elderly people in India may prefer to communicate in regional languages rather than English, but existing alert systems may not adequately support such interactions. There is therefore a need for a low-cost, privacy-preserving system that detects falls without wearable

devices or cameras and responds to a detected fall in the person's own language under predictable safety rules.

4.3 OBJECTS OF THE INVENTION

The principal objects of the present invention are:

1. to provide a non-intrusive fall-detection system that needs no camera and no wearable device;
2. to detect falls from amplitude information of a single low-cost Wi-Fi receiver, without multi-antenna phase processing;
3. to provide a compact neural network that runs on low-cost edge hardware and discriminates falls from periodic activities such as walking and running;
4. to provide an event-driven voice interaction that confirms the event with the person in a preferred language, interprets varied replies, asks follow-up questions when a reply is unclear, and handles the absence of a reply;
5. to ensure that a rule-based safety controller takes critical escalation decisions and does not depend solely on the output of a language model;
6. to notify a remote caregiver automatically, with context, and to track delivery and acknowledgment of the notification; and
7. to retain contextual information about the person and past incidents for continuity across interactions, and to support routine check-ins and reminders using the same framework.

4.4 NOVELTY OF THE INVENTION

The novelty of the present invention lies in the following features and in their combination. For each feature, the limitation of the patent documents [1] to [4] is stated first, followed by the manner in which the present invention overcomes it. As far as the applicant is aware, these features are not disclosed together in [1] to [4].

1. United States Patent No. US 10,531,817 B2 [1] (Peking University) detects falls from the CSI of at least two receiving antennas, using the phase difference between them, manually designed features and a one-class Support Vector Machine. It therefore requires multi-antenna hardware and phase processing, which low-cost commodity microcontrollers with a single antenna and unreliable raw phase information may not

provide, and its hand-designed features may struggle to distinguish periodic activities such as walking or running from falls. It only detects the fall and does not interact with the person.

How the present invention addresses this: The sensing node of the present invention needs only a single receiving antenna and uses the amplitude of 52 fixed active subcarriers, with no phase information, so that a low-cost commodity microcontroller is sufficient. Instead of hand-designed features and a one-class classifier, a compact neural network (FallGuardNet, 21,571 trainable parameters, trained by knowledge distillation) learns the features itself, and a periodicity score and a stillness score computed inside the network help it to separate falls from periodic activities such as walking or running. After a confirmed fall the system does not stop at detection: it starts a spoken interaction with the person through the agentic conversational layer.

2. Chinese Patent No. CN 112346050 B [2] (Tsinghua University) estimates the speed of a human body from Wi-Fi CSI and supplies it to a trained fall detection model. It outputs only a detection result, without any interaction with the person or escalation procedure, and is not described as executing on low-cost edge hardware.

   How the present invention addresses this: In the present invention the trained network runs locally on a low-cost edge processing unit, such as a Raspberry Pi 5, so that no separate server is needed for classification. The detection result is not the end of the process but an event that is sent to the agentic conversational layer. That layer creates an incident, speaks to the person, interprets the reply and, under a safety policy, escalates to a caregiver, so that the system provides both detection and a complete escalation procedure.

3. United States Patent Application Publication No. US 2024/0112559 A1 [3] (Google LLC) detects a fall from radar reflections, announces it through a speaker, listens for a spoken reply and alerts an emergency contact. It requires dedicated radar hardware rather than commodity Wi-Fi and, as far as the applicant is aware, does not describe a rule-based escalation decision kept separate from the interpretation of the reply by a language model, clarifying questions for unclear replies, tracking of delivery and acknowledgement of the notification, or retention of context for later interactions.

How the present invention addresses this: The present invention senses the fall from commodity Wi-Fi signals, so that no radar hardware is required. In one embodiment the language model supplies only a recognised intent, and a separate rule-based safety-policy controller decides whether to confirm, ask again, wait or escalate, so that critical escalation never depends solely on a generated response. An unclear reply leads to a clarifying question and a missing reply leads to a configured timeout and then escalation, and the incident workflow has distinct outcomes for each case. The notification to the caregiver is tracked for delivery and acknowledgement, and an incident store retains the incident and the context of the person for later interactions and for scheduled check-ins. The spoken interaction is held in the preferred language of the person, which is taken from a profile, and the speech and language processing can be executed locally on the edge unit.

4. United States Patent No. US 8,217,795 B2 [4] detects a fall from the accelerations measured by a wearable sensor and sends an alert to a console or call centre. It requires the person to wear and charge a device and does not interact with the person.

   How the present invention addresses this: The present invention is contactless: the person wears and charges nothing, because the fall is sensed from the changes that movement causes in Wi-Fi signals, and no camera is used, which preserves privacy. The system also talks to the person, asking whether help is needed and acting on the reply, and it alerts the caregiver, with the context of the incident, through a messaging service and a caregiver application, instead of sending a bare alert to a console or call centre.

4.5 DETAILED DESCRIPTION OF THE INVENTION

The invention is described below with reference to the accompanying drawings. The embodiments are illustrative and not limiting, and like reference numerals denote like parts throughout.

1. System overview (FIG. 1 and FIG. 2)

Referring to FIG. 1, the fall-detection system (100) comprises a Wi-Fi sensing node (101) having a single receiving antenna (102); a Wi-Fi access point (103); an edge processing unit (104), which in the described embodiment is a Raspberry Pi 5 single-board computer; a voice interface (105) comprising a microphone (105a) and a speaker (105b); optionally a cloud database (106); a caregiver application (107) running on a caregiver device (108); and a

messaging service (109). The edge processing unit (104) hosts a frame receiver (110), a windowing and normalization module (111), an inference module (112), a decision module (113), an agentic conversational layer (114) and an escalation module (115). The modules (110) to (115) are implemented as software instructions stored in a non-transitory memory of the edge processing unit (104) and executed by its processor.

Referring to FIG. 2, the sensing node (101) is placed in a room (202) in which a resident (201) lives. Radio waves exchanged between the access point (103) and the sensing node (101) travel along several paths (203) that reflect off walls, furniture, and the resident's body. Resident movement, including a fall, changes these paths and therefore the CSI measured by the sensing node. The resident is monitored without any wearable device and without capturing images, so privacy is preserved.

2. Sensing node (FIG. 3)

The sensing node (101) comprises an ESP32-class microcontroller with a single receiving antenna (102). In step S301, it connects as a station to the access point (103) and disables Wi-Fi power saving. In step S302, it enables CSI reporting with legacy and high-throughput long training fields and channel filtering, and registers a callback invoked for every received data frame (step S303).

In step S304 the callback reads the raw CSI buffer, which contains 64 complex samples, each consisting of an in-phase component I and a quadrature component Q. In step S305, it computes, for each sample, an amplitude equal to the square root of (I squared plus Q squared). In step S30,6, it keeps only 52 active subcarriers of an IEEE 802.11n 20 MHz channel: bins 38 to 63 (the 26 negative-index subcarriers) and bins 1 to 26 (the 26 positive-index subcarriers), skipping the DC bin and the guard bands. When the hardware flags the first word of the CSI buffer as invalid, it substitutes the corrupted bin 1 with the amplitude of the adjacent bin 2. No phase value, and no phase difference between antennas, is computed or used.

In step S307, the callback places a frame, consisting of a timestamp, a frame number, and the 52 amplitudes, on a bounded queue and performs no network input or output itself, so the Wi-Fi receive path is never blocked. In step S308, a separate writer task dequeues the frame, and in step S309 transmits it as a text datagram over the User Datagram Protocol (UDP) to the edge processing unit (104). The process repeats for every received frame.

3. Windowing and normalization (FIG. 4)

The frame receiver (110) receives each datagram (step S401), validates that it carries exactly 52 amplitude values, and appends the frame to a buffer (step S402). In one embodiment, frames arrive at about 100 Hz (100 packets per second, ~10 ms interval). When the buffer holds 950 consecutive frames (step S403), the windowing and normalization module (111) forms a 950 × 52 matrix. Non-finite values are replaced by zero (step S404). The matrix is then normalized by a global per-window z-score, equal to (x − mean) divided by (standard deviation + ε), computed over the entire matrix (step S405). This makes the classification independent of absolute signal strength and of the distance between the sensing node and the person. The normalized matrix is supplied to the inference module (112), which executes the neural network through an ONNX runtime.

4. Neural network FallGuardNet (FIG. 5)

The network receives the normalized window (501) of 950 × 52 values and transposes it (502) into 52 channels over 950 time steps. The network comprises:

1. a **fine branch** (503) of three depthwise-separable one-dimensional convolution layers with kernel size 3 and dilations 1, 2 and 4, which captures sharp impact transients;
2. a **medium branch** (504) of three such layers with kernel size 7 and dilations 1, 2 and 4, which captures motion patterns such as a gait cycle;
3. a **coarse branch** (505) of two such layers with kernel size 15 and dilations 1 and 2, which captures slow posture-change trends;
4. a **temporal alignment and concatenation** stage (506) that aligns the outputs of the three branches by interpolation and concatenates them, followed by a **fusion layer** (507) comprising a 1 × 1 convolution, batch normalization and a GELU activation, which reduces the channels;
5. a **squeeze-and-excitation channel attention** block (508) that weights the fused channels;
6. a **temporal attention pooling** (509) that computes a softmax-weighted sum over time, together with a **max pooling** (510) over time;
7. a **periodicity score** module (511) that computes the Pearson correlation of the per-frame signal energy of the window with the same energy delayed by a lag of 100 frames, which is high for walking or running and low for a fall;

8. a **stillness score** module (512) that computes one minus the ratio of the motion energy (the mean squared first difference over time) in the last 20 % of the window to the motion energy in the first 80 % of the window, which is high when the signal becomes calm after an abrupt event, as after a fall; and
9. a **classifier head** (514) comprising layer normalization, a linear layer, a GELU activation, dropout and a final linear layer, which receives the concatenation (513) of the attention-pooled features, the max-pooled features, the periodicity score and the stillness score, and produces two output logits (515), one for non-fall and one for fall.

The network computes the periodicity and stillness scores alongside learned features and combines them with the classifier head, so the decision does not depend on a single hand-set energy-ratio threshold. In one embodiment, the channel width is 32 and the network has 21,571 trainable parameters, which is small enough for execution on the edge processing unit (104) in real time.

5. Training by knowledge distillation (FIG. 6)

Labelled CSI amplitude windows (601) are augmented (602) and sampled by a class-weighted sampler (603). A larger teacher network (604), in one embodiment an 18-layer residual convolutional network of about 11 million parameters, is first trained on the windows with class-weighted cross-entropy and is then held fixed. The student network (606), having the architecture of FallGuardNet described above, is trained with a combined loss (610) equal to α times a distillation loss (608) plus $(1 - \alpha)$ times a hard-label loss (609). The distillation loss (608) is the Kullback–Leibler divergence between the temperature-softened outputs (605) of the teacher and the temperature-softened outputs (607) of the student, scaled by the square of the temperature; the hard-label loss (609) is the class-weighted cross-entropy between the student outputs and the true labels. In one embodiment α is 0.7 and the temperature is 4. An optimiser (611) updates the student weights.

In one embodiment the training data are CSI amplitude recordings of seven activities (arm wave, fall, jump, run, squat, turn and walk), each of 950 frames by 52 subcarriers, mapped to a binary label (fall or non-fall). Fall windows are augmented by time reversal and circular time shift, and all windows are augmented by random time shift, amplitude scaling by a factor between 0.85 and 1.15, and low-level Gaussian noise. The trained student network (612) is exported in ONNX format (613) and deployed on the edge processing unit (104).

6. Decision module (FIG. 4)

The inference module (112) produces two logits (step S406), which are converted by a softmax function to a fall probability p (step S407). The decision module (113) compares p with a confirmed-fall threshold T2 (step S408) and a possible-fall threshold T1 (step S411). In one embodiment T1 is 0.30 and T2 is 0.65. If p is at or above T2, the state is CONFIRMED FALL (step S409) and an event is sent once to the agentic conversational layer (114) (step S410). If p is below T2 but at or above T1, or is somewhat lower but accompanied by an elevated signal-motion measure computed from the same window, the state is POSSIBLE FALL (step S412). Otherwise the state is NON-FALL (step S413) and a latch that prevents repeated triggering is cleared. The signal-motion measures are computed from median-filtered, frame-normalised CSI; they are used to display activity and to widen the possible-fall band, and never to override the confirmed-fall decision. A confirmed fall latches the alert state for a fixed period. The event sent to the conversational layer includes its classification and relevant details, such as the fall probability and the window index.

7. Agentic conversational layer (FIG. 7)

The agentic conversational layer (114) is a software layer executed by the processor of the edge processing unit (104). It begins when it receives an event from the decision module (113), or a scheduled task from a scheduler (710), and it manages the subsequent interaction, interpretation, state tracking and escalation. The decision module (113) remains responsible for CSI-based sensing and event classification. The layer comprises:

10. An **event interface** (701) through which the decision module sends the event; in one embodiment an endpoint of a FastAPI application, and in other embodiments an MQTT or serial interface;
11. An **incident manager** (702) that creates an incident record for each event and updates its current state;
12. An **incident store** (703), being persistent storage, in one embodiment a SQLite database, holding the preferred language, contact details and preferences of the person, previous incident records and schedules;
13. A **conversation state machine** (704) that tracks the current incident and conversation state, for example awaiting reply, clarifying, no response, escalated and resolved, and determines what happens next; in one embodiment implemented in Python, and in

another embodiment by a graph-based orchestration framework with checkpointing, such as LangGraph, where the workflow is complex enough to benefit from it;

14. A **speech synthesis engine** (705), in one embodiment a local text-to-speech engine such as Piper, that converts a prompt in the preferred language into audio played through the speaker (105b); for a small fixed set of prompts, prerecorded audio may be used instead;
15. A **voice activity detector** (706), in one embodiment a neural detector such as Silero VAD and in another an energy-threshold detector, that distinguishes speech from silence or non-speech audio captured by the microphone (105a), so that recording starts and stops appropriately and long silent recordings are not processed;
16. A **speech-to-text engine** (707), in one embodiment faster-whisper, that converts the speech segment into text in the language of the person;
17. An **interpretation module** (708) that uses a language model, in one embodiment a local model served through Ollama such as Qwen2.5-1.5B-Instruct, to map varied expressions to a recognised intent, so that replies such as "I'm fine", "I'm okay" or "Theek hoon" may indicate that the person reports being safe, and replies such as "I'm not able to get up" and "I need help" may both indicate that assistance is needed; a keyword-based multilingual matcher may be used as a fallback or as a cross-check, and the recognised intent is subject to validation;
18. A **safety-policy controller** (709), being rule-based, that evaluates the incident state and the recognised intent and decides whether to follow a confirmation policy, ask again, wait, or escalate; and
19. Optionally a **scheduler** (710) that initiates scheduled check-ins and medication reminders.

An output of the safety-policy controller (709) that calls for escalation is passed to the escalation module (115), which notifies the caregiver through the messaging service (109). The speech-to-text engine, the language model and the speech synthesis engine may all be executed locally on the edge processing unit, so that audio and transcripts need not be sent to an external service, which supports privacy and operation without an internet connection for the interaction itself.

8. Incident workflow (FIG. 8)

In step S801 the decision module (113) sends an event, comprising its classification and relevant details, to the conversational layer (114), which receives it through the event interface (701). In step S802 the incident manager (702) creates an incident in the incident store (703) and retrieves context, such as the preferred language, contact details and previous incident records of the person, where available. In step S803 the conversation state machine (704) starts the incident workflow.

In step S804 the speech synthesis engine (705) speaks a prompt in the preferred language through the speaker (105b), for example asking whether the person is okay and whether the person needs help. In step S805 the microphone (105a) captures the response. In step S806 the voice activity detector (706) detects the speech. In step S807 the speech-to-text engine (707) transcribes the audio. In step S808 the interpretation module (708) interprets the response and produces a recognised intent.

In step S809 the safety-policy controller (709) evaluates the incident state and the recognised intent and selects one of four outcomes:

20. **Clearly okay.** The workflow follows a configured confirmation policy (step S810) and then saves the outcome (step S817). In one embodiment the policy records the incident as resolved and resumes monitoring; in another it requests a second confirmation or sends an informational message to the caregiver.
21. **Unclear response.** The state machine asks a clarifying question (step S811) and returns to capture of a new response (step S805). In one embodiment the number of clarifying questions is limited by configuration, after which the response is treated as unusable.
22. **No usable response.** The state machine waits for a configured timeout (step S812). If a response is then received (step S813), the workflow returns to step S805. If there is still no response, the workflow proceeds to escalation (step S814).
23. **Needs help or reports pain.** The workflow proceeds directly to escalation (step S814).

In step S814 caregiver escalation is triggered. In step S815 the escalation module (115) sends a caregiver notification and in step S816 it tracks delivery and acknowledgement of the notification. In step S817 the outcome is saved and the incident is updated in the incident

store (703). A scheduled check-in or medication reminder (step S818) may initiate the same workflow from step S802.

9. Safety rules and escalation

The safety-policy controller (709) applies predefined rules. If the person requests help, reports being unsafe or in pain, or fails to respond in accordance with the configured policy, the controller initiates the escalation procedure. The language model contributes only a recognised intent, which is one input to the rules; critical escalation decisions do not depend solely on a response generated by the language model, and uncertainty, silence or failure of the microphone or of a speech component leads, after the configured clarification and timeout steps, to escalation rather than to dismissal of the event.

The escalation module (115) sends, through the messaging service (109), a notification to the caregiver device (108) and, in one embodiment, to the caregiver application (107). The notification states the time, the resident, a summary of the interaction, the fall probability and the change in signal energy. The module tracks whether the notification was delivered and acknowledged and records the result in the incident. In one embodiment the incident records and device status, including a periodic heartbeat, are also synchronised to a cloud database (106) read by the caregiver application, so that the caregiver can see that the sensor is online.

10. Persistent context and routine care

The outcome of each incident, relevant interaction details and any follow-up action are recorded in the incident store (703). This information supports continuity in later interactions, subject to privacy and data-retention rules, such as a configurable retention period. Outside emergency events, the same conversational framework can initiate scheduled check-ins and medication reminders: the scheduler (710) retrieves the relevant schedule from the incident store, the conversational layer delivers a voice prompt, records and interprets the response, and updates the interaction state.

11. Example implementation of the conversational layer

Table 1 lists, by way of example, components of the conversational layer and technologies that may be used to implement them. The technologies listed are examples of implementation choices; any equivalent technology may be used, and the final selection reflects what is integrated and tested in a given product.

Component	Example technology	Purpose and usage
Application logic	Python	Coordinates event handling, speech processing, conversation flow, memory and escalation decisions
Event communication interface (701)	FastAPI endpoint; alternatively MQTT or serial	Receives event details from the fall-detection module
Voice activity detection (706)	Silero VAD (optional)	Identifies portions of audio containing speech, reducing unnecessary recognition processing
Speech-to-text (707)	faster-whisper	Converts the spoken response into text
Interpretation (708)	Local language model served by Ollama, for example Qwen2.5-1.5B-Instruct	Interprets varied natural-language responses and supports context-aware replies
Conversation and safety control (704, 709)	Python state machine and rule-based logic; LangGraph optionally	Tracks the incident, manages conversation states, handles unclear or absent responses and applies escalation rules
Speech synthesis (705)	Piper	Converts replies into audio played through the speaker
Persistent storage (703)	MongoDB	Stores language, contacts, incident records, schedules and preferences

Table 1

12. Experimental results (FIG. 9)

In a preliminary evaluation of the fall classification on a held-out test split comprising 60 non-fall and 10 fall recordings, the trained student network produced 60 true negatives, 0 false positives, 8 true positives and 2 false negatives (FIG. 9(a)), corresponding to an accuracy of 97.1 %, a precision of 100 %, a recall of 80 %, a specificity of 100 % and an F1 score of 0.889. Table 2 compares the network with other models evaluated on the same split (FIG. 9(b) and FIG. 9(c)).

Model	F1 score	Specificity
MLP	0.455	0.883
CNN	0.562	0.783
GRU	0.476	0.900

Model	F1 score	Specificity
LSTM	0.321	0.383
Transformer	0.667	0.867
MobileNetV3	0.750	0.917
FallGuardNet, first version	0.667	0.867
FallGuardNet (present invention)	**0.889**	**1.000**

Table 2

13. Further embodiments

The sensing node may be any Wi-Fi chip that exposes CSI. A different number of subcarriers, window length, kernel sizes, dilations, channel width or thresholds may be used. In the described embodiment the edge processing unit is a Raspberry Pi 5; the term edge processing unit also includes any equivalent processing unit that performs the same functions, and the conversational layer may be executed on the edge processing unit or on another local processing device connected to it. The speech-recognition, language-model and speech-synthesis components may be local or remote services, and the language model may be omitted in an embodiment in which a keyword-based matcher interprets the response. The cloud database and the messaging service may be any equivalent services. The foregoing description of embodiments is illustrative, and modifications that do not depart from the scope of the invention will be apparent to a person skilled in the art.

ABSTRACT OF THE INVENTION

The invention provides a fall-detection system comprising: a Wi-Fi sensing node, such as an ESP32 microcontroller, that extracts amplitude values of a fixed set of active OFDM subcarriers from received CSI frames and streams them over a network; and an edge processing unit, such as a Raspberry Pi 5 single-board computer, that buffers the frames into a window of fixed length, normalizes the window, and classifies it with a multi-resolution temporal neural network (FallGuardNet) into fall or non-fall. A decision module applies two probability thresholds to produce a possible-fall state or a confirmed-fall state and sends an event to an agentic conversational layer.

The agentic conversational layer, on receiving the event, creates an incident record, retrieves context such as the preferred language and contact details of the person, speaks to the person in that language, captures the reply with the help of voice activity detection, converts it to

text, and interprets it with a language model. A state machine tracks the incident state, asks a clarifying question when the reply is unclear, and moves to a no-response state when no reply is received within a configured period. A separate rule-based safety-policy controller evaluates the incident state and the recognized intent and decides whether to follow a confirmation policy, ask again, wait, or escalate. An escalation module notifies a caregiver, tracks delivery and acknowledgment, and saves the outcome to persistent storage.

Unlike multi-antenna phase-difference approaches, the sensing node uses a single receiving antenna and amplitude only. The neural network combines three parallel dilated depthwise-separable convolution branches with different kernel sizes, channel attention, temporal attention pooling, and two signal-derived scalar features: a periodicity score and a stillness score. It is trained by knowledge distillation from a larger teacher network so that it is small enough to run on the edge unit. The language model supplies only a recognized intent to the safety-policy controller, so that critical escalation does not depend solely on a generated response.

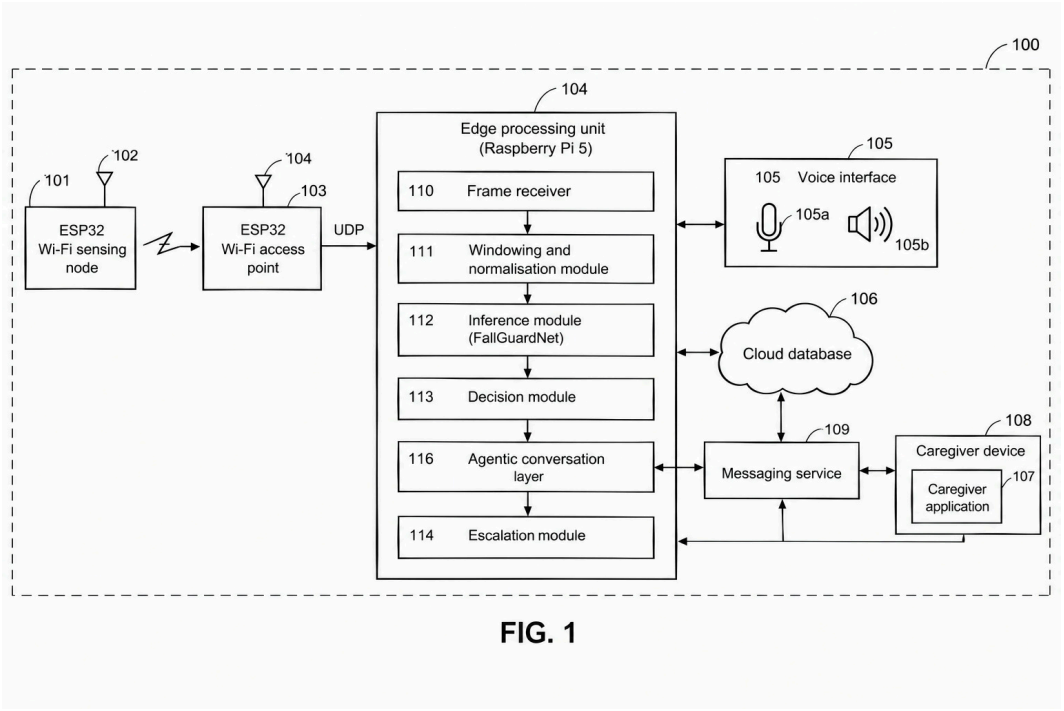


FIG. 1

FIG. 1 is a block diagram of the fall-detection system according to an embodiment of the invention;

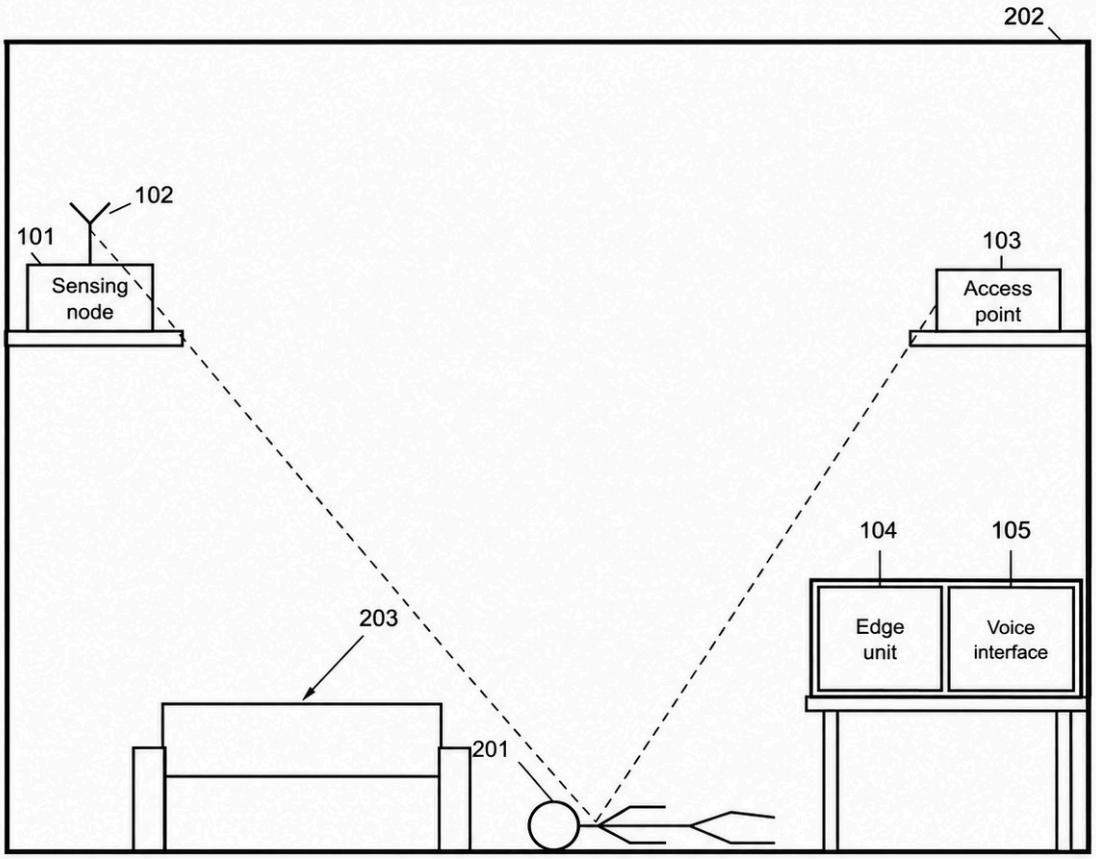


FIG. 2

FIG. 2 is a schematic view of an example deployment of the system in a room;

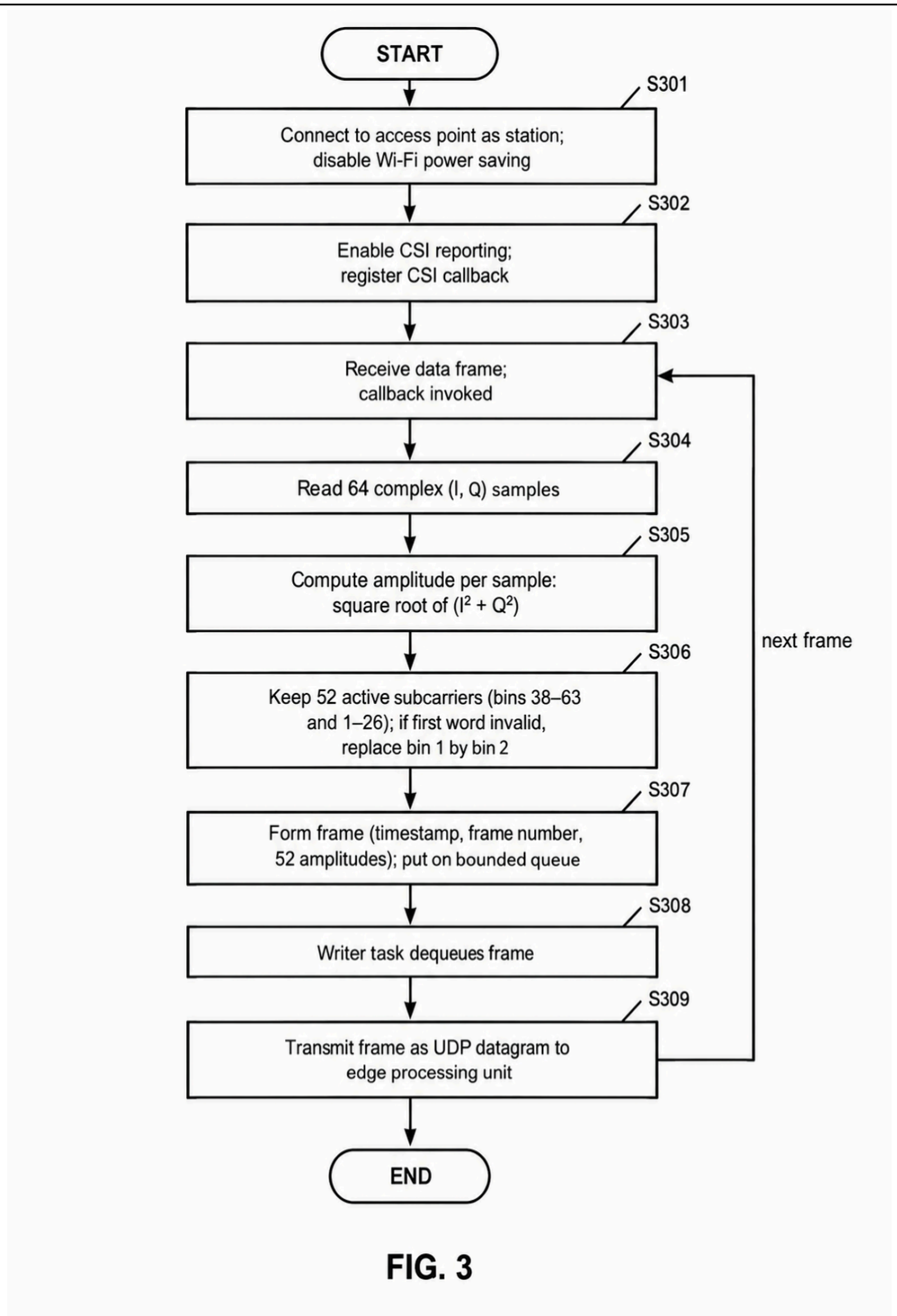


FIG. 3

FIG. 3 is a flow chart of CSI acquisition and subcarrier selection at the sensing node;

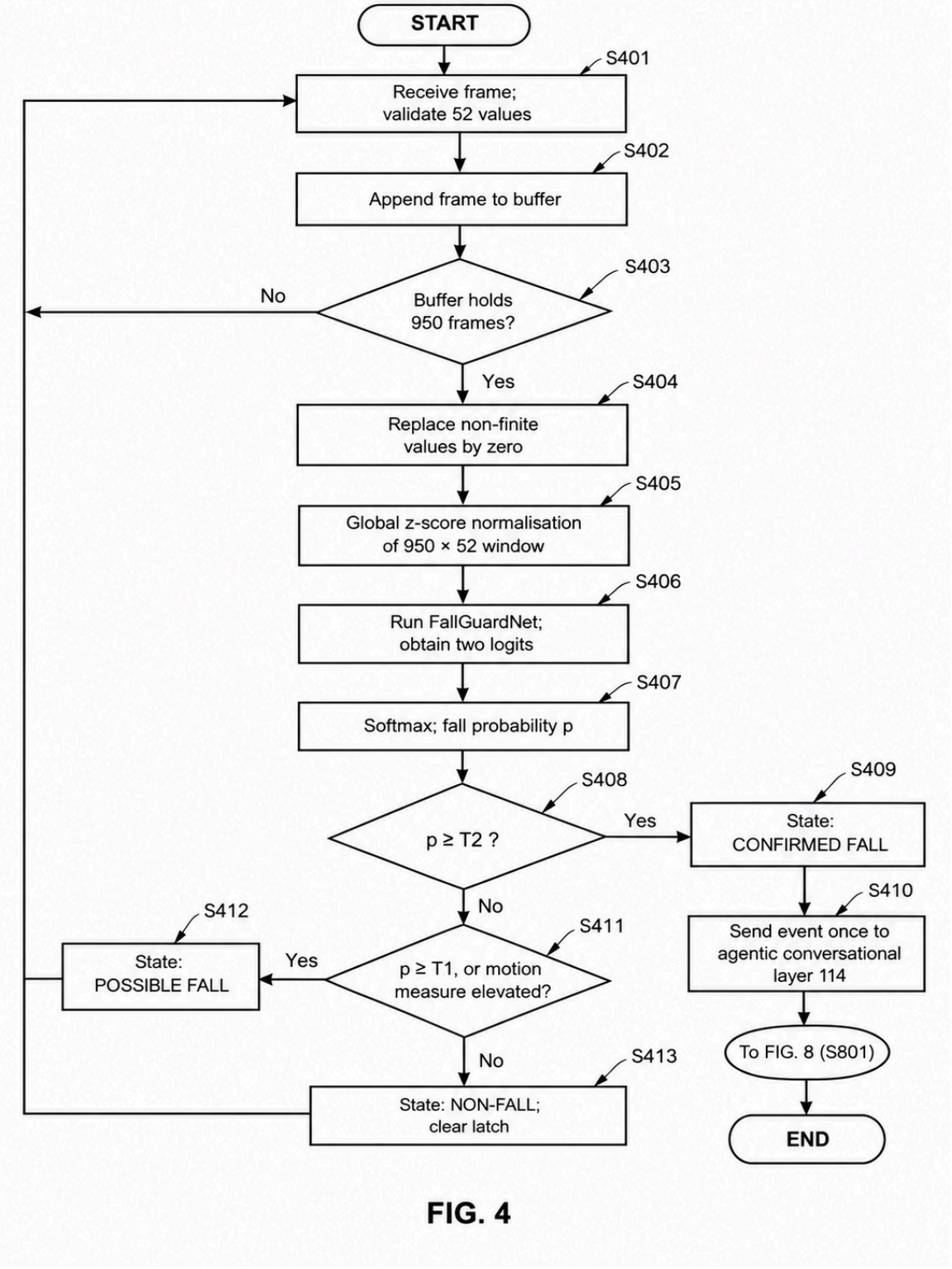


FIG. 4

FIG. 4 is a flow chart of window processing, thresholding and state determination at the edge unit;

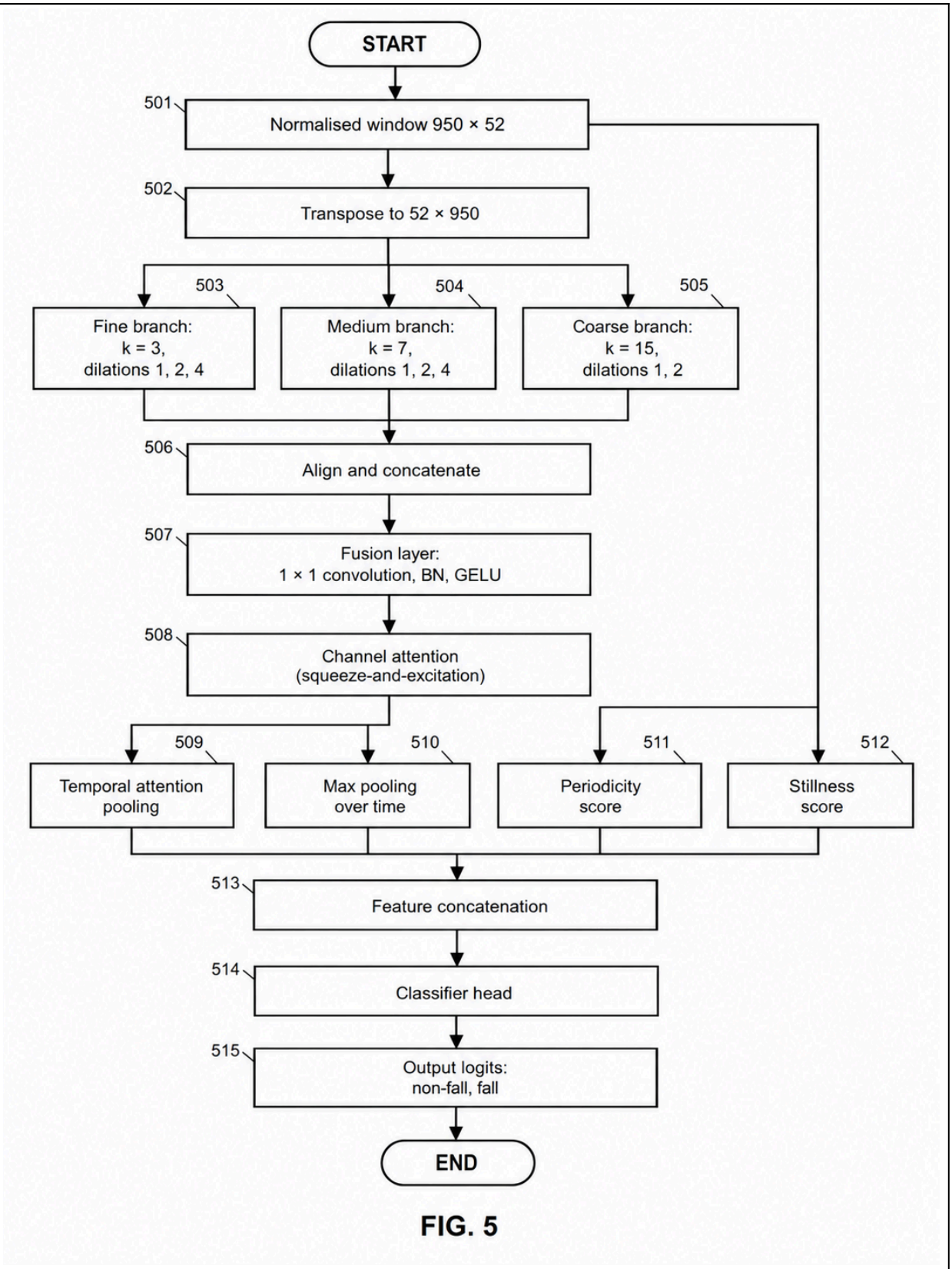


FIG. 5

FIG. 5 is a block diagram of the architecture of the FallGuardNet neural network;

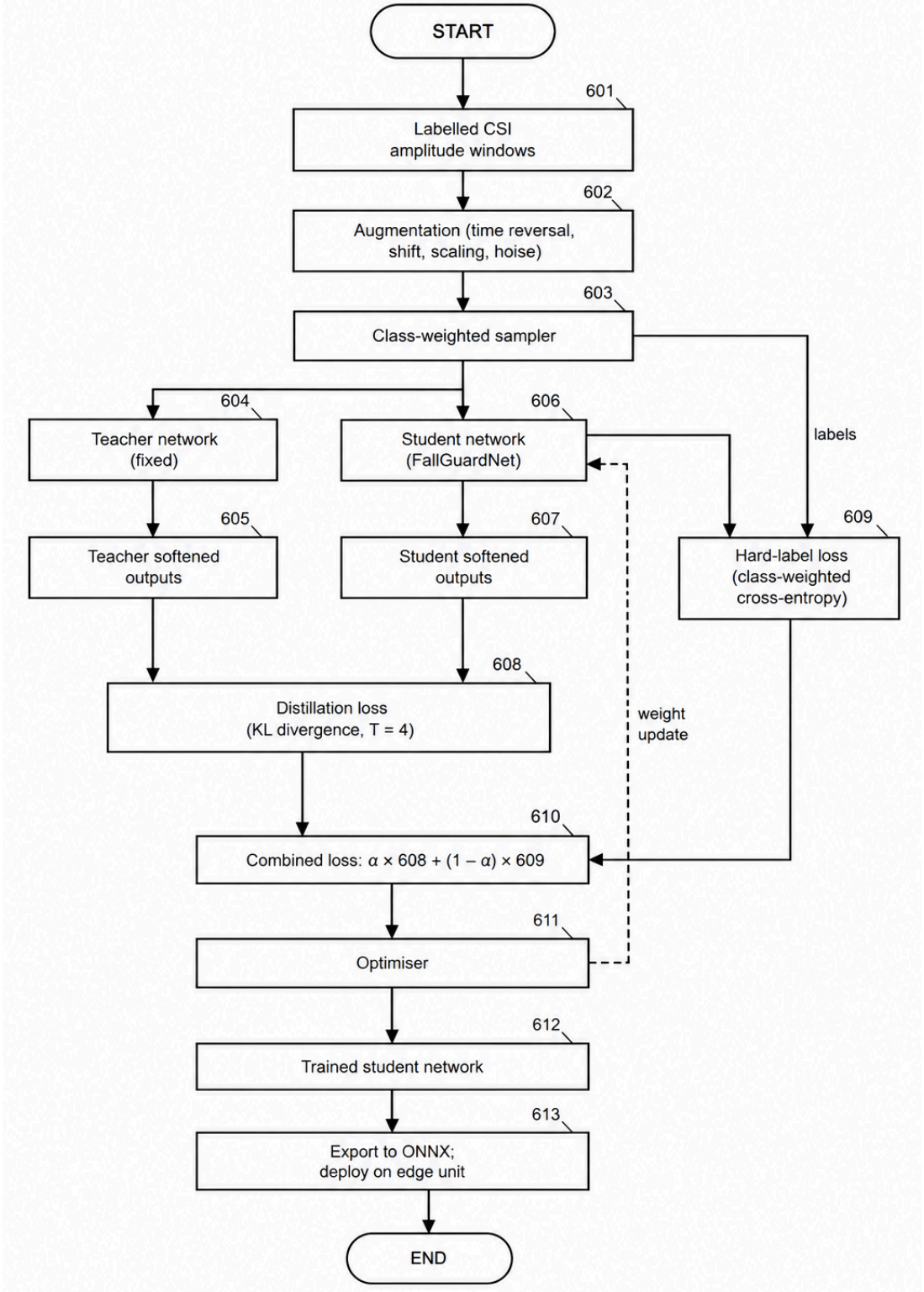


FIG. 6

FIG. 6 is a block diagram of training of the student network by knowledge distillation;

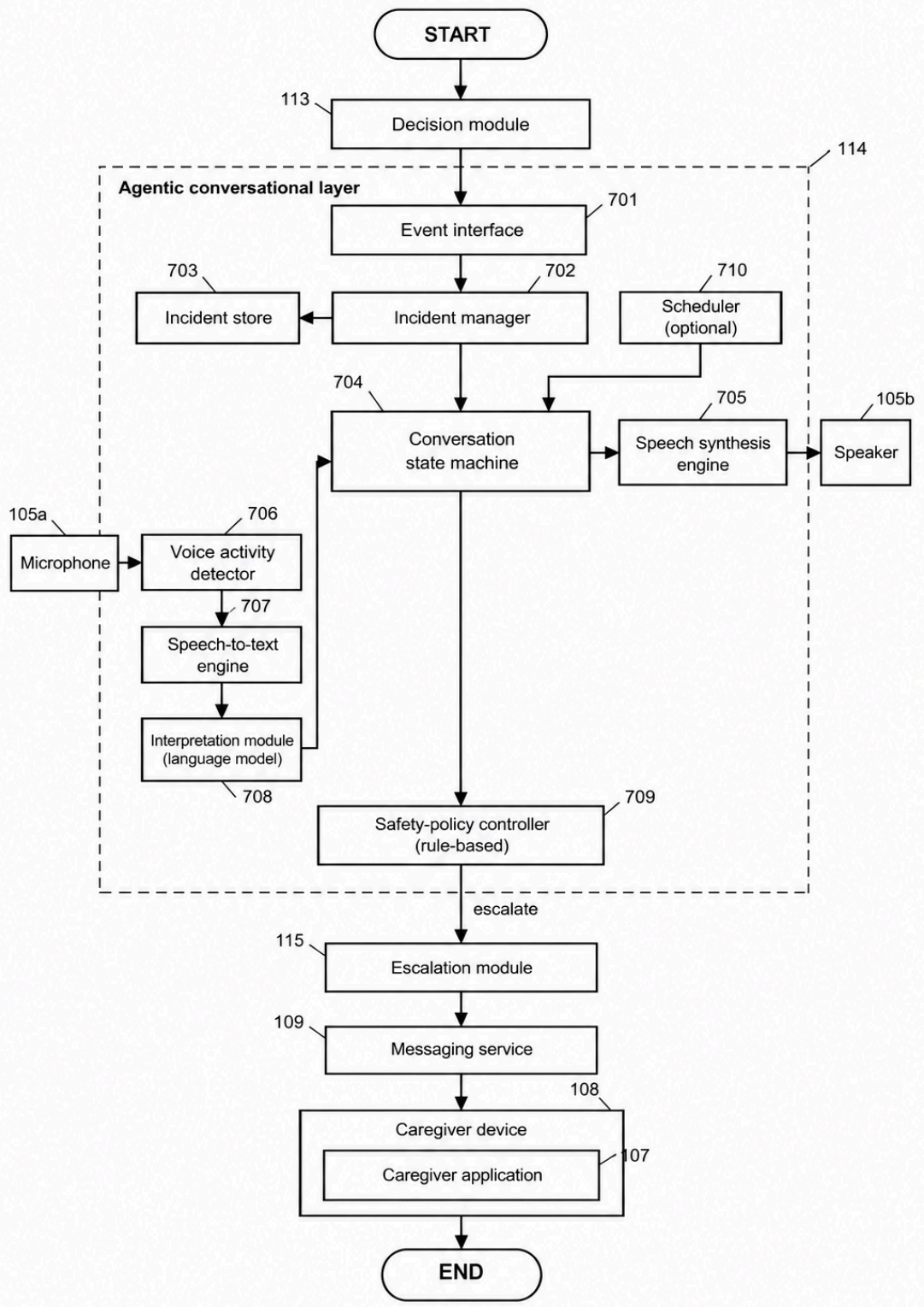


FIG. 7

FIG. 7 is a block diagram of the agentic conversational layer;

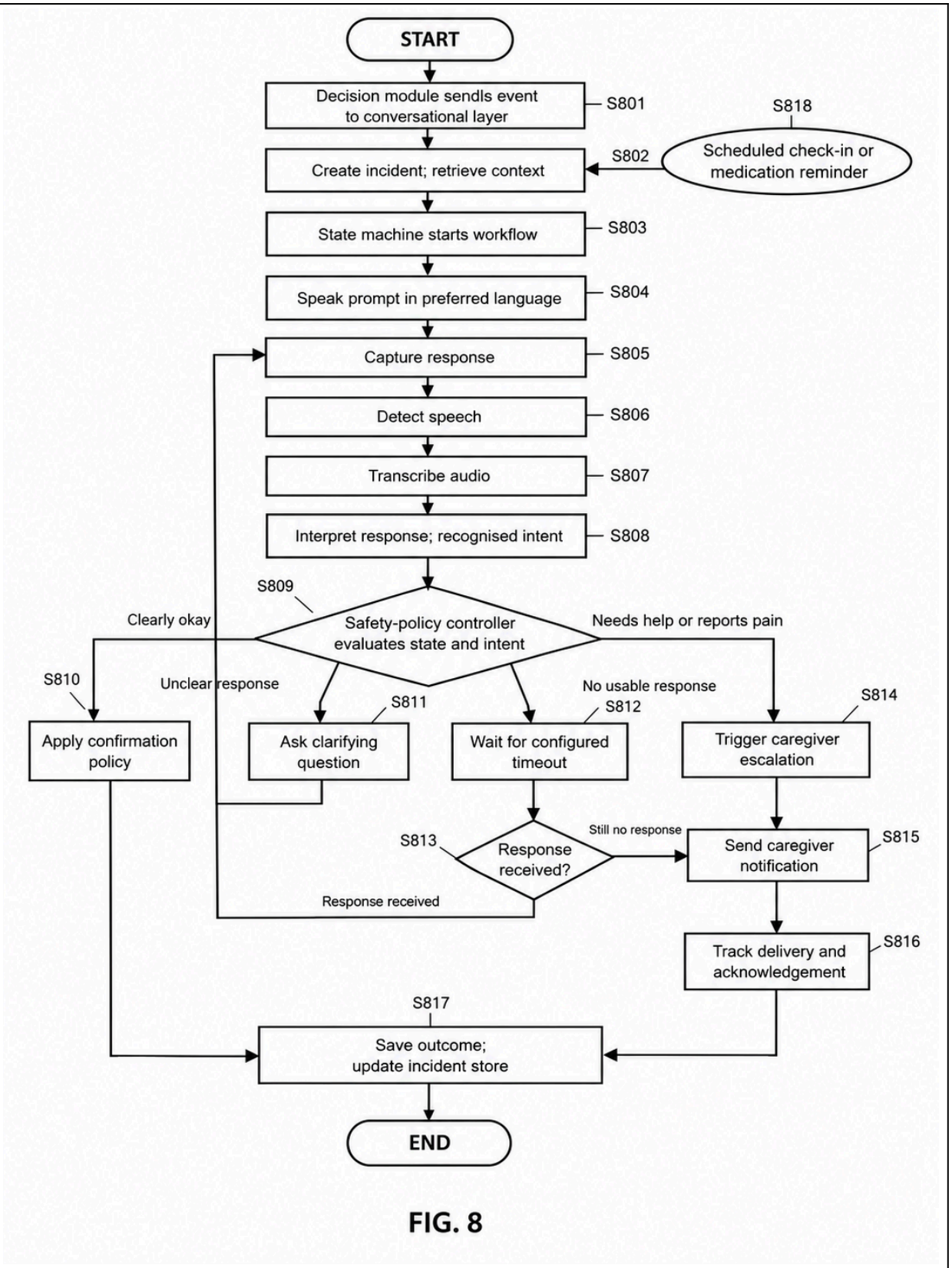


FIG. 8

FIG. 8 is a flow chart of the incident workflow performed by the agentic conversational layer

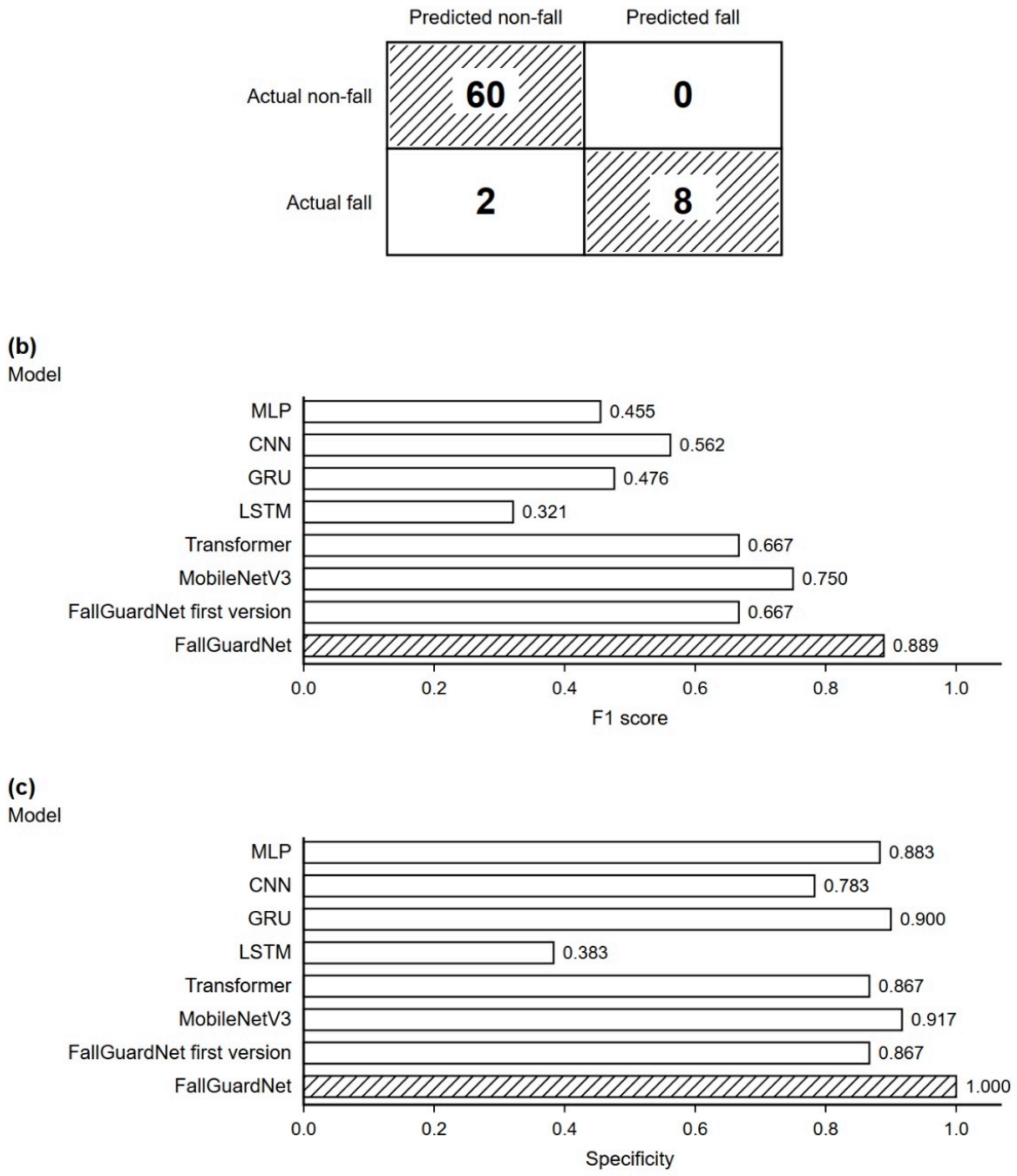


FIG. 9

FIG. 9 shows experimental results: (a) a confusion matrix, (b) a comparison of F1 score and (c) a comparison of specificity with other models.

REFERENCES

The following patent documents are referred to in this specification.

[1] United States Patent No. US 10,531,817 B2, “Fall Detection Method and System,” Peking University.

[2] Chinese Patent No. CN 112346050 B, “Fall detection method and system based on Wi-Fi device,” Tsinghua University, granted 28 June 2024.

[3] United States Patent Application Publication No. US 2024/0112559 A1, “Privacy-Preserving Radar-Based Fall Monitoring,” Google LLC, published 4 April 2024.

[4] United States Patent No. US 8,217,795 B2, “Method and system for fall detection,” J. Carlton-Foss, granted 10 July 2012.